Exp.no : 20 **PANDAS LIBRARY**

AIM:

Experiment to understand pandas library use cases in Data science.

PYTHON CODE:

```
import pandas as pd

# Create a dataset
data = {
   'Name': ['Arun', 'Bala', 'Cathy', 'David', 'Eva'],
   'Age': [20, 21, 19, 22, 20],
   'Marks': [85, 78, 92, 67, 88]
}
df = pd.DataFrame(data)
# Display the dataset
print("Original Dataset:")
print(df)
# Display first five rows
print("\nFirst Five Rows:")
print(df.head())
# Display dataset information
print("\nDataset Information:")
print(df.info())
# Calculate statistical summary
print("\nStatistical Summary:")
print(df.describe())

# Filter students with marks above 80
print("\nStudents Scoring Above 80:")
print(df[df['Marks'] > 80])

# Calculate average marks
print("\nAverage Marks:", df['Marks'].mean())
```

OUTPUT:

```
Original Dataset:
    Name  Age  Marks
0   Arun   20     85
```

```
1   Bala   21     78
2  Cathy   19     92
3  David   22     67
4    Eva   20     88
First Five Rows:
    Name  Age  Marks
0   Arun   20     85
1   Bala   21     78
2  Cathy   19     92
3  David   22     67
4    Eva   20     88
Dataset Information:<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Name    5 non-null      object
 1   Age     5 non-null      int64
 2   Marks   5 non-null      int64
dtypes: int64(2), object(1)
memory usage: 172.0+ bytes
None
Statistical Summary:
           Age      Marks
count   5.000000   5.000000
mean   20.400000  82.000000
std     1.140175   9.823441
min    19.000000  67.000000
25%    20.000000  78.000000
50%    20.000000  85.000000
75%    21.000000  88.000000
max    22.000000  92.000000
Students Scoring Above 80:
    Name  Age  Marks
0   Arun   20     85
2  Cathy   19     92
4    Eva   20     88
Average Marks: 82.0
```

RESULT:

The code is executed successfully.